ML Model for Predictive Maintenance in Industrial Systems

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1. Introduction

Industrial machines operate continuously under varying operational conditions, making them susceptible to wear and unexpected failures. Such failures can lead to significant downtime, increased maintenance costs, and safety risks. Traditional maintenance strategies such as reactive maintenance or scheduled preventive maintenance often fail to detect early signs of equipment degradation.

With the growing availability of industrial sensor data and Industrial IoT technologies, machine learning provides an effective solution for predictive maintenance. By analysing patterns in historical machine data, ML models can identify early indicators of faults and help predict failures before they occur, allowing maintenance to be scheduled proactively.

2. Problem Statement

The goal of this project is to develop a machine learning model that can predict potential equipment failures using historical sensor data. Machine parameters such as temperature, vibration, pressure, and rotational speed often show measurable changes prior to failure.

The challenge lies in identifying meaningful patterns in large volumes of sensor data and distinguishing between normal operational variations and early fault indicators. This project aims to train ML models that can classify machine health conditions and estimate the likelihood of failure.

3. Dataset Description

The project will use a publicly available predictive maintenance dataset containing sensor measurements and machine operational parameters. Typical features include temperature, vibration levels, rotational speed, and machine usage cycles. Each data instance represents a machine's operational state along with a label indicating whether a failure occurred.

One suitable dataset for this project is the AI4I 2020 Predictive Maintenance Dataset, which simulates industrial machine behavior with multiple sensor readings and labeled failure types.

4. Methodology

The project will follow a standard machine learning workflow.

First, the dataset will undergo preprocessing steps such as handling missing values, normalization, and exploratory data analysis. Relevant features will be selected to represent machine behaviour effectively.

Next, several machine learning models will be implemented and compared, including Logistic Regression, Decision Trees, Random Forest, and Support Vector Machines (SVM). The dataset will be divided into training and testing sets to evaluate model performance using metrics such as accuracy, precision, recall, and F1-score.

5. Expected Outcomes

The expected outcome of this project is a machine learning-based predictive maintenance system capable of identifying potential equipment failures in advance. By comparing multiple ML models, the study aims to determine the most effective approach for predicting machine faults. The results are expected to demonstrate how predictive maintenance can improve system reliability and reduce industrial downtime.

6. References

- Mobley, R. K. An Introduction to Predictive Maintenance. Elsevier, 2002.
- Zhang, W., Yang, D., & Wang, H. (2019). Data-driven methods for predictive maintenance of industrial equipment. IEEE Systems Journal.
- AI4I 2020 Predictive Maintenance Dataset – Kaggle.